Notes on Computational Fluid Dynamics: General Principles

5.14 Descent methods

The Gauss-Seidel method, introduced in Sec. 5.2, provides convergent solutions for many problems in CFD. It is most effective when a modest reduction in residual is required, *e.g.* as part of a steady-state solution described in Sec. 5.12.

When the Gauss-Seidel method requires a lot of sweeps (*e.g.* over 10) to converge to a suitable tolerance, alternative methods may be more efficient. *Descent methods* provide alternative matrix solvers that are often used in CFD.

Descent methods represent the equations which are being solved as a *minimisation* problem. It is demonstrated below using a matrix equation of the form $\mathbf{A} \cdot \mathbf{\Psi} = \mathbf{b}$ with 2 values

$$\begin{bmatrix} 6 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} \Psi_1 \\ \Psi_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}. \tag{5.27}$$

The minimisation presents the equation in *quadratic form*; in matrix notation, it is a *scalar* function of the form

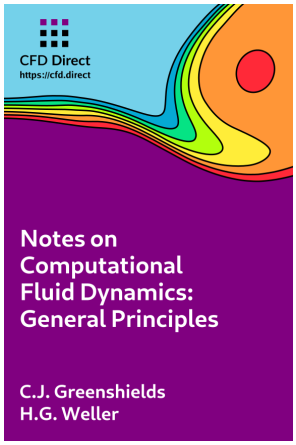
$$f(\mathbf{\Psi}) = \frac{1}{2}\mathbf{\Psi} \cdot \mathbf{A} \cdot \mathbf{\Psi} - \mathbf{b} \cdot \mathbf{\Psi} \tag{5.28}$$

The quadratic form of Eq. (5.27), is:

$$f(\mathbf{\Psi}) = 3\Psi_1^2 + 2\Psi_2^2 - 2\Psi_1\Psi_2 - \Psi_1 - 3\Psi_2 \tag{5.29}$$

The gradient of $f(\mathbf{\Psi})$ is:

$$\frac{\partial f(\mathbf{\Psi})}{\partial \mathbf{\Psi}} = \begin{bmatrix} \partial f/\partial \Psi_1 \\ \partial f/\partial \Psi_2 \end{bmatrix} = \begin{bmatrix} 6\Psi_1 - 2\Psi_2 - 1 \\ -2\Psi_1 + 4\Psi_2 - 3 \end{bmatrix} \tag{5.30}$$



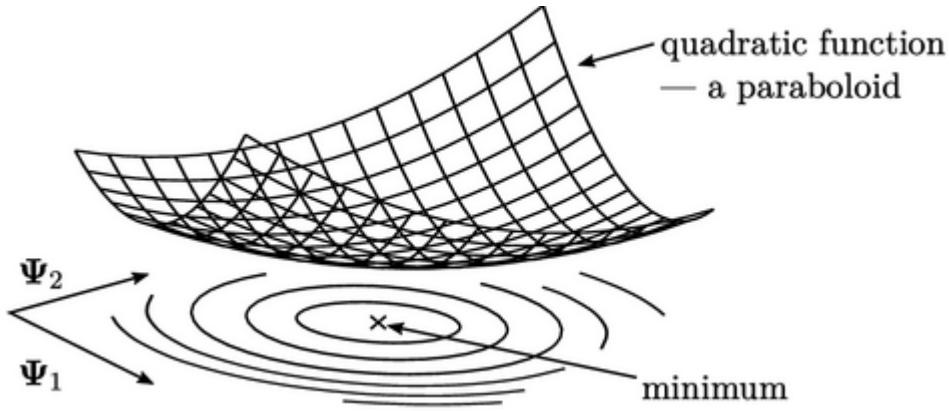
Buy the Printed Book

available at amazon

Amazon.com (US)
Amazon.co.uk (UK)
Amazon.com.au (Australia)
Amazon.ca (Canada)
Amazon.com.mx (Mexico)
Amazon.de (Germany)
Amazon.fr (France)
Amazon.es (Spain)
Amazon.it (Italy)
Amazon.nl (Netherlands)

Critically, $\partial f(\boldsymbol{\Psi})/\partial \boldsymbol{\Psi} = \mathbf{A} \bullet \boldsymbol{\Psi} - \mathbf{b} = -\mathbf{r}$, *i.e.* the negative of the residual vector Eq. (5.10), as verified by the model example.

Equating the gradient to zero, $\partial f(\boldsymbol{\Psi})/\partial \boldsymbol{\Psi} = \mathbf{0}$, corresponds to a *minimum in the quadratic function*. At the same time, it is the *solution* to $\mathbf{A} \bullet \boldsymbol{\Psi} = \mathbf{b}$. The method is therefore concerned with finding the minimum of the quadratic form efficiently.



For this method to work, the quadratic form must have a minimum, which requires that $\mathbf{A}$ is *symmetric* and *positive-definite*. A positive-definite matrix is hard to visualise, but for a 2-value function it ensures the quadratic function is a paraboloid.

Diagonal dominance is the convergence condition for the Gauss-Seidel method, discussed in Sec. 5.3. Importantly, a symmetric matrix that is diagonally dominant, and has positive diagonal coefficients, is also positive-definite.

Matrix operations

The '$\bullet$' operation between two single-column matrices, *e.g.* $\mathbf{b} \bullet \boldsymbol{\Psi}$, in Eq. (5.28) is represented in other texts using matrix notation by a transpose $[\mathbf{b}]^{\mathrm{T}} [\boldsymbol{\Psi}]$. For the example in Eq. (5.27), it is:

$$\mathbf{b} \bullet \boldsymbol{\Psi} = [\mathbf{b}]^{\mathrm{T}} [\boldsymbol{\Psi}] = \begin{bmatrix} 1 & 3 \end{bmatrix} \begin{bmatrix} \Psi_1 \\ \Psi_2 \end{bmatrix} = \Psi_1 + 3\Psi_2 \tag{5.31}$$

Chris Greenshields Henry Weller 11th April 2022 Notes on CFD

Amazon.se (Sweden)
Amazon.pl (Poland)
Amazon.jp (Japan)
Amazon.com.tr (Türkiye)

OpenFOAM Training

16 Sep Cologne, Germany
23 Sep Virtual, Europe
07 Oct Virtual, Americas
14 Oct Virtual, Europe
28 Oct Chicago, USA
05 Nov Virtual, Americas

Copyright © 2015-2023 CFD Direct Ltd Contact Us Twitter LinkedIn YouTube Website Terms of Use Privacy